

CodeWeaver comparison with CRUST-Bench

On all 100 exact CRUST-Bench subjects, 300/300 CodeWeaver cells compiled and 165/300 passed every fixed project test. Mean project success was 55.00% (sample SD 1.00 pp). The paper's single-shot and three-round repair settings are preserved as references; CodeWeaver's multi-stage five-repair/three-parity protocol is not relabeled as those settings.

Method

This exact-subject re-analysis imports the published CodeWeaver campaign's independently restored CRUST interfaces and fixed tests. All three terminal outcomes are retained; no best-of-three selection is used.

Validity boundary

CRUST-Bench's Table 4 reports pass rates under single-shot, compiler repair, test repair, and an adapted SWE-agent. CodeWeaver uses a different architecture and larger repair budget. Comparison is descriptive, not a controlled model ablation.

Exact CodeWeaver measurements

Run	Build	Pass all	Fixed tests	Test rate
CodeWeaver rep 1	100/100	56/100 (56.00%)	500/623	80.26%
CodeWeaver rep 2	100/100	55/100 (55.00%)	507/623	81.38%
CodeWeaver rep 3	100/100	54/100 (54.00%)	464/623	74.48%
CodeWeaver mean	100.00%	55.00% +/- 1.00 pp	three independent repetitions	95% t-CI +/- 2.48 pp

Published CRUST-Bench Table 4

System	Base build	Base test	Compiler-repair build	Compiler-repair test	Test-repair build	Test-repair test
OpenAI o3	35.00%	19.00%	68.00%	31.00%	63.00%	48.00%
Claude Opus 4	43.00%	22.00%	78.00%	29.00%	65.00%	40.00%
OpenAI o1	32.00%	15.00%	69.00%	28.00%	54.00%	37.00%
Claude 3.7	26.00%	13.00%	54.00%	23.00%	49.00%	32.00%
Claude 3.5	26.00%	11.00%	49.00%	21.00%	38.00%	24.00%
o1-mini	19.00%	9.00%	47.00%	16.00%	27.00%	21.00%
GPT-4o	18.00%	7.00%	52.00%	18.00%	42.00%	22.00%
Gemini 1.5 Pro	11.00%	3.00%	35.00%	11.00%	30.00%	14.00%
Virtuoso (Distilled Deepseek V3)	2.00%	2.00%	21.00%	6.00%	10.00%	6.00%
Deepseek-Coder-32B	1.00%	0.00%	2.00%	0.00%	2.00%	0.00%

System	Base build	Base test	Compiler-repair build	Compiler-repair test	Test-repair build	Test-repair test
QwQ-32B-Preview	1.00%	0.00%	1.00%	0.00%	1.00%	0.00%
Qwen-2.5-Coder-32B	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Adapted SWE-agent (Claude-3.7)	41.00%	32.00%	N/A	N/A	N/A	N/A